



Advanced Time Series Analysis and Forecasting

Seminar 6: State space models and Bayesian filtering



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Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **APF**: Auxiliary Particle Filter
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BN**: Beveridge–Nelson (decomposition)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CEE**: Central and Eastern Europe
- ▣ **DFM**: Dynamic Factor Model
- ▣ **DK**: Durbin–Koopman (simulation smoother, 2002)
- ▣ **ESS**: Effective Sample Size
- ▣ **FFBS**: Forward Filtering, Backward Sampling
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP**: Gross Domestic Product

Acronyms used in this material (2/2)

- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **IG**: inverse gamma (distribution)
- ▣ **IMA**: Integrated Moving Average (model)
- ▣ **KSC**: Kim–Shephard–Chib (1998), mixture sampler for stochastic volatility
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MA**: Moving Average
- ▣ **ML**: Machine Learning
- ▣ **MNZ**: Morley–Nelson–Zivot (2003)
- ▣ **PMMH**: Particle Marginal Metropolis–Hastings
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **SCA**: Seasonally and Calendar Adjusted
- ▣ **SV**: Stochastic Volatility
- ▣ **TSA**: Time Series Analysis
- ▣ **TVP**: Time-Varying Parameter(s)
- ▣ **UC**: Unobserved Components (model)
- ▣ **UC-SV**: Unobserved Components model with Stochastic Volatility (Stock–Watson 2007)
- ▣ **US**: United States
- ▣ **VAT**: Value Added Tax

Today's question and route

- **Question:** how do we estimate what we cannot observe (a level, a trend, a volatility, a gap), and how do we tell a real signal from the noise of the estimator?
 - ▶ this seminar comes **before** Lecture 6: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: the exact diffuse filter by hand, the concentrated likelihood, backward sampling, the Gibbs block, SV moments, one particle-filter step, the APF, Beveridge–Nelson
 - ▶ Part B: Romanian trend growth; TVP regressions for Hungary and Poland; SV of EUR/RON by Gibbs and by the particle filter; the real-time Romanian output gap
 - ▶ Part C: is Romanian trend inflation anchored? An AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	the exact diffuse Kalman filter by hand; the concentrated likelihood	Solved, Proposed	A1
A3, A4	one backward-sampling step; the inverse-gamma Gibbs block	Solved, Proposed	A3
A5, A6	moments of the SV model; the QML measurement equation	Solved, Proposed	A5
A7, A8	one bootstrap particle-filter step; auxiliary weights with an outlier	Solved, Proposed	A7
A9, A10	the Beveridge–Nelson cycle; local level, IMA(1,1) and BN	Solved, Proposed	A9
B1, B2	Romanian trend growth; TVP regressions for Hungary and Poland	Solved, Proposed	B1
B3, B4	SV of EUR/RON: Gibbs, then the particle filter	Solved, Proposed	B3
B5	the Romanian output gap in real time	Proposed	B1, A10
C1, C2	is Romanian trend inflation anchored?; what is wrong in an AI answer?	Proposed	B3, A1–A10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
Romania: real GDP (SCA)	Eurostat namq_10_gdp	quarterly	2000–2026
HICP: Romania, Hungary, Poland, euro area	Eurostat prc_hicp_minr	monthly	2001–2026
EUR/RON reference rate	BNR	daily	2016–2026

- In the notebook: `kalman_filter`, `kalman_smoother`, `sim_smoother_dk`, `ffbs`, `sv_gibbs`, `pf_sv`, `fit_llt`, `fit_tvp`, `fit_smooth_trend`, `ucsv_gibbs`; no account or key is needed

Prerequisites for Today (1/19): the state space model

- Model (TSA, Chapter 10): observations are a noisy linear function of a hidden state that follows a linear dynamic

$$y_t = Z_t \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H); \quad \alpha_{t+1} = T \alpha_t + R \eta_t, \quad \eta_t \sim N(0, Q)$$

- Notation

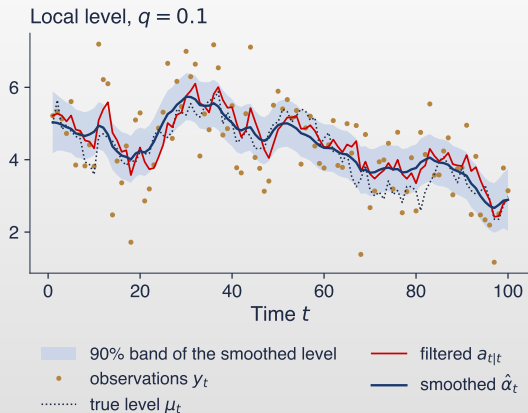
- ▶ y_t : the observation; α_t : the unobserved state; Z_t : measurement matrix; T : transition matrix; R : selects the states with shocks
- ▶ H : variance of the measurement noise ε_t ; Q : covariance of the state shocks η_t ; n : the sample length

- Three estimates of the state, $Y_t = (y_1, \dots, y_t)$

- ▶ predicted $a_t = \mathbb{E}(\alpha_t \mid Y_{t-1})$, $P_t = \text{Var}(\alpha_t \mid Y_{t-1})$; filtered $a_{t|t} = \mathbb{E}(\alpha_t \mid Y_t)$, $P_{t|t}$
- ▶ smoothed $\hat{\alpha}_t = \mathbb{E}(\alpha_t \mid Y_n)$, $V_t = \text{Var}(\alpha_t \mid Y_n)$: uses the whole sample

- Local level: $Z = T = R = 1$, $y_t = \mu_t + \varepsilon_t$, $\mu_{t+1} = \mu_t + \eta_t$; signal-to-noise ratio $q = \sigma_\eta^2 / \sigma_\varepsilon^2$

Prerequisites for Today (2/19): filtered and smoothed level



- ▣ Simulated local level, $n = 100$, $\sigma_\varepsilon^2 = 1$, $q = 0.1$
 - ▶ dots: the data; dotted line: the true level μ_t
- ▣ Filtered $a_{t|t}$ (red): uses data up to t only
 - ▶ what an analyst knows in real time; it reacts late
- ▣ Smoothed $\hat{a}_t$ (blue): uses all n observations
 - ▶ smoother and closer to μ_t ; band $\hat{a}_t \pm 1.645\sqrt{V_t}$
 - ▶ at $t = n$ the two coincide

Prerequisites for Today (3/19): the Kalman filter

- Prediction error and its variance

$$v_t = y_t - Z_t a_t, \quad F_t = Z_t P_t Z_t' + H$$

- ▶ v_t : the new information in y_t (the innovation), independent of Y_{t-1}

- Update and prediction, with the Kalman gain $K_t = T P_t Z_t' F_t^{-1}$

$$a_{t|t} = a_t + P_t Z_t' F_t^{-1} v_t, \quad a_{t+1} = T a_t + K_t v_t, \quad P_{t+1} = T P_t (T - K_t Z_t)' + R Q R'$$

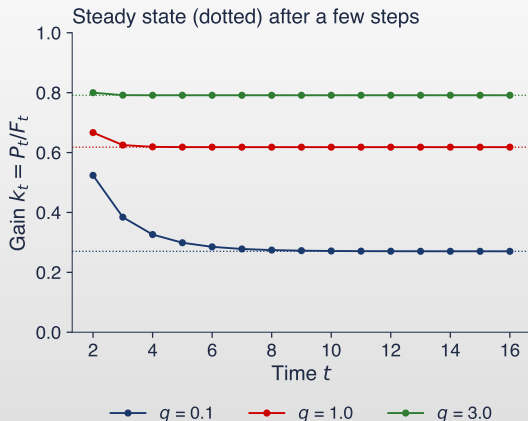
- ▶ K_t : how much of the surprise v_t moves the next state prediction; large when P_t is large relative to H
- ▶ local level: $a_{t+1} = a_t + k_t v_t$, $k_t = P_t / (P_t + \sigma_\varepsilon^2)$: an exponentially weighted average of the data

- Log-likelihood by the prediction error decomposition, a by-product of the filter

$$\ln L = -\frac{1}{2} \sum_{t=1}^n (\ln 2\pi + \ln F_t + v_t^2 / F_t)$$

- ▶ maximised numerically over the variances in H and Q

Prerequisites for Today (4/19): the gain and the smoother



- Local level, units of σ_ε^2 : $P_{t+1} = P_t - P_t^2 / (P_t + 1) + q$
 - ▶ P_t and k_t converge after a few steps (dotted lines): the steady state
 - ▶ a larger q : a larger gain, the level follows the data closely

- Smoother, backward from $t = n$

$$\hat{\alpha}_t = a_{t|t} + G_t(\hat{\alpha}_{t+1} - T a_{t|t})$$

- ▶ $G_t = P_{t|t} T' P_{t+1}^{-1}$: how much the next state tells about the current one
- ▶ $V_t = P_{t|t} + G_t(V_{t+1} - P_{t+1})G_t'$

Prerequisites for Today (5/19): the exact diffuse start

- Nonstationary states (random walks) have no stationary variance

$$P_1 = \kappa P_\infty + P_*, \quad \kappa \rightarrow \infty$$

- ▶ P_∞ : marks the diffuse elements; P_* : covariance of the others; $F_{\infty,t} = Z_t P_{\infty,t} Z_t'$ and $F_{*,t}$: the diffuse and finite parts of F_t

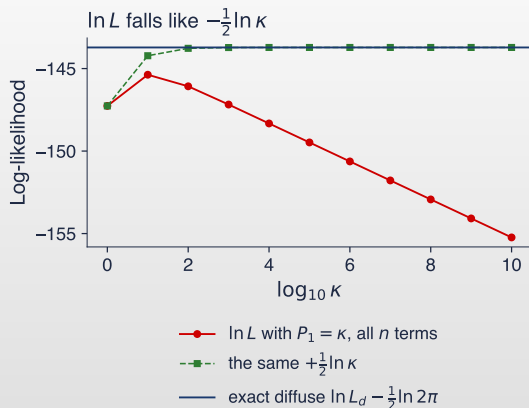
- While $F_{\infty,t} > 0$, two gains [Koopman \(1997\)](#)

$$K^{(0)} = TP_\infty Z' / F_\infty, \quad K^{(1)} = T(P_* Z' - P_\infty Z' F_* / F_\infty) / F_\infty$$

- ▶ $a_{t+1} = Ta_t + K^{(0)} v_t$, $P_{\infty,t+1} = TP_\infty L^{(0)'}$, $P_{*,t+1} = TP_\infty L^{(1)'} + TP_* L^{(0)'} + RQR'$
- ▶ $L^{(0)} = T - K^{(0)}Z$, $L^{(1)} = -K^{(1)}Z$; after d diffuse steps $P_\infty = 0$ and the ordinary filter takes over

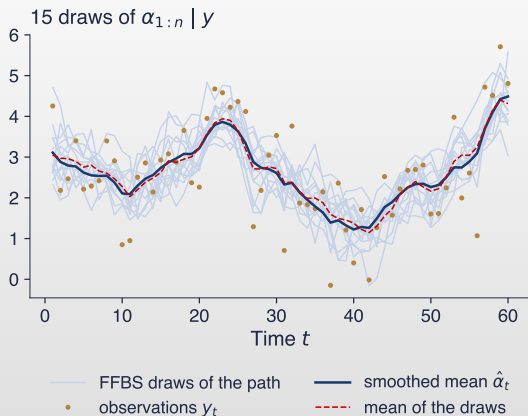
- Diffuse log-likelihood: a diffuse step contributes $-\frac{1}{2}(\ln 2\pi + \ln F_{\infty,t})$, the others the usual term

- ▶ scale concentrated out: if H , Q are proportional to σ^2 , $\hat{\sigma}^2 = \frac{1}{n-d} \sum_{t>d} v_t^2 / F_t^*$, $F_t^* = F_t / \sigma^2$

Prerequisites for Today (6/19): a large κ against the exact diffuse start

- Simulated local level, $n = 80$, $q = 0.5$
 - ▶ red: $a_1 = 0$, $P_1 = \kappa$ and all n terms of $\ln L$
- The first term contains $-\frac{1}{2}\ln(\kappa + \sigma_\varepsilon^2)$
 - ▶ $\ln L \rightarrow -\infty$ as κ grows; round-off errors grow too
 - ▶ adding back $\frac{1}{2}\ln \kappa$ (green) recovers the exact diffuse value (blue)
- Use the exact diffuse filter and drop the d diffuse terms

Prerequisites for Today (7/19): sampling the states



- FFBS [Carter and Kohn \(1994\)](#): forward filter, backward sampling

- ▶ draw $\alpha_n \sim N(a_{n|n}, P_{n|n})$

- ▶ then, for $t = n - 1, \dots, 1$: $\alpha_t | \alpha_{t+1} \sim N(m_t, S_t)$

$$m_t = a_{t|t} + G_t(\alpha_{t+1} - T a_{t|t}), \quad S_t = P_{t|t} - G_t T P_{t|t}$$

- ▶ $G_t = P_{t|t} T' (T P_{t|t} T' + R Q R')^{-1}$

- Durbin–Koopman [Durbin and Koopman \(2002\)](#): simulate (α^+, y^+) , smooth once

$$\tilde{\alpha} = \alpha^+ + \hat{\alpha}(y) - \hat{\alpha}(y^+)$$

- ▶ $\hat{\alpha}(\cdot)$: the smoothed mean; $\tilde{\alpha}$: an exact draw of the whole path

- Chart: the draws scatter around $\hat{\alpha}_t$; their mean is $\hat{\alpha}_t$

Prerequisites for Today (8/19): Gibbs blocks for the variances

- ▣ Inverse gamma $IG(a, b)$: $\sigma^2 \sim IG(a, b)$ if $1/\sigma^2 \sim \text{Gamma}$ with shape a and rate b

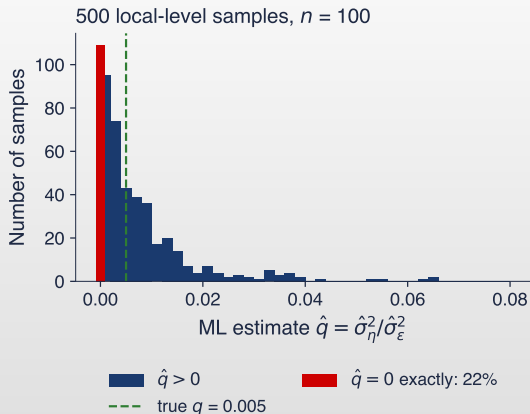
$$p(\sigma^2) \propto (\sigma^2)^{-a-1} e^{-b/\sigma^2}, \quad \mathbb{E}\sigma^2 = \frac{b}{a-1} \quad (a > 1)$$

- ▶ a : shape; b : scale of σ^2 (the rate of $1/\sigma^2$); the density is zero at $\sigma^2 = 0$
- ▣ Gibbs block: prior $IG(a_0, b_0)$, m shocks e_t implied by the current state draw

$$\sigma^2 \mid e \sim IG\left(a_0 + \frac{m}{2}, b_0 + \frac{1}{2} \sum_t e_t^2\right)$$

- ▶ local level: $e_t = y_t - \mu_t$ for σ_ε^2 ($m = n$), $e_t = \Delta\mu_t$ for σ_η^2 ($m = n - 1$)
- ▶ the sampler alternates: states given variances (FFBS), variances given states
- ▣ Inefficiency factor $1 + 2 \sum_{k \geq 1} \rho_k$
 - ▶ ρ_k : autocorrelation of the chain at lag k ; the number of correlated draws worth one independent draw

Prerequisites for Today (9/19): variances at the boundary



- Pile-up [Shephard and Harvey \(1990\)](#): the ML estimate of a variance can be exactly 0
 - ▶ 500 local-level samples, $n = 100$, true $q = 0.005$
 - ▶ 22% of the estimates are exactly $\hat{q} = 0$
- LR test of $\sigma_\eta^2 = 0$: the null is on the boundary
 - ▶ $LR = 2(\ln L_1 - \ln L_0) \geq 0$; standard χ_1^2 is wrong
 - ▶ rule of thumb: $p = \frac{1}{2}P(\chi_1^2 > LR)$ (mixture $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$)
 - ▶ random-walk alternatives: nonstandard; use a parametric bootstrap

Prerequisites for Today (10/19): local trends and time-varying coefficients

▣ Local linear trend

$$y_t = \mu_t + \varepsilon_t, \quad \mu_{t+1} = \mu_t + \nu_t + \xi_t, \quad \nu_{t+1} = \nu_t + \zeta_t$$

- ▶ ν_t : the slope; ξ_t, ζ_t : level and slope shocks with variances $\sigma_\xi^2, \sigma_\zeta^2$; two diffuse states
- ▶ $y_t = 100 \ln$ of a quarterly series: $4\nu_t$ is the annualised trend growth, in %

▣ TVP (time-varying parameter) regression

$$y_t = c_t + b_t x_t + D_t' \gamma + \varepsilon_t, \quad c_{t+1} = c_t + \eta_t^c, \quad b_{t+1} = b_t + \eta_t^b$$

- ▶ state $\alpha_t = (c_t, b_t, \gamma')'$, $Z_t = (1, x_t, D_t')$; D_t : seasonal dummies, γ : constant states
- ▶ $\sigma_b^2 = \text{Var}(\eta_t^b) = 0$: a constant slope; the smoothed b_t shows when the link changed

Prerequisites for Today (11/19): stochastic volatility

- SV model: the log-variance is a latent AR(1) with its own shock

$$y_t = e^{h_t/2} \epsilon_t, \quad h_t = \mu + \phi(h_{t-1} - \mu) + \sigma_\eta \eta_t, \quad \epsilon_t, \eta_t \sim N(0, 1)$$

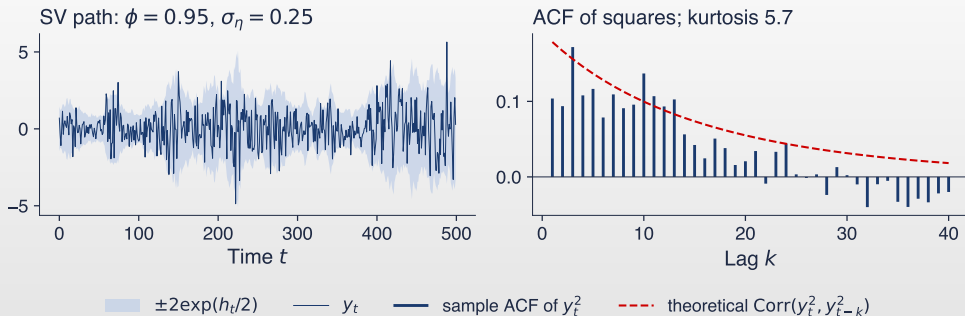
- ▶ h_t : log-variance; μ : its mean; ϕ : persistence; σ_η : volatility of volatility
- ▶ half-life of a shock to h_t : $\ln 0.5 / \ln \phi$ periods

- Moments, with $\sigma_h^2 = \sigma_\eta^2 / (1 - \phi^2)$ the variance of h_t

$$\text{kurtosis} = 3e^{\sigma_h^2}, \quad \text{Corr}(y_t^2, y_{t-k}^2) = \frac{e^{\sigma_h^2 \phi^k} - 1}{3e^{\sigma_h^2} - 1}$$

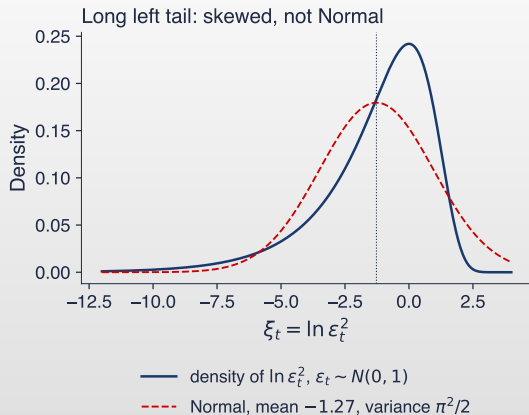
- ▶ kurtosis above 3 and positive, slowly decaying autocorrelation of y_t^2 : volatility clustering
- GARCH(1,1) [Bollerslev \(1986\)](#), for comparison: $\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ the variance is known one step ahead; in SV it has its own shock, so the likelihood is an integral over $h_{1:n}$

Prerequisites for Today (12/19): a simulated SV path



- $\phi = 0.95$, $\sigma_\eta = 0.25$: $\sigma_h^2 = 0.0625/0.0975 = 0.64$, kurtosis $3e^{0.64} = 5.7$, half-life $\ln 0.5 / \ln 0.95 = 13.5$ periods
- Left: calm and turbulent spells inside the band $\pm 2e^{h_t/2}$; right: the ACF of y_t^2 starts low (0.18) and decays like ϕ^k

Prerequisites for Today (13/19): the linearised SV model



- Square and take logs

$$\ln y_t^2 = h_t + \xi_t, \quad \xi_t = \ln \varepsilon_t^2$$

- ▶ $\mathbb{E}\xi_t = -1.2704$, $\text{Var}\xi_t = \pi^2/2 = 4.93$
- ▶ linear in h_t : a state space model

- QML Harvey, Ruiz and Shephard (1994): the Kalman filter as if ξ_t were Normal

- ▶ best linear predictor, but ξ_t is skewed (chart)

- KSC Kim, Shephard and Chib (1998): a mixture of seven Normals

$$p(\xi_t) \approx \sum_{j=1}^7 q_j N(\xi_t; m_j, v_j)$$

- ▶ q_j, m_j, v_j : fixed weights, means, variances
- ▶ given the component s_t , h is Gaussian: FFBS applies

Prerequisites for Today (14/19): particle filters

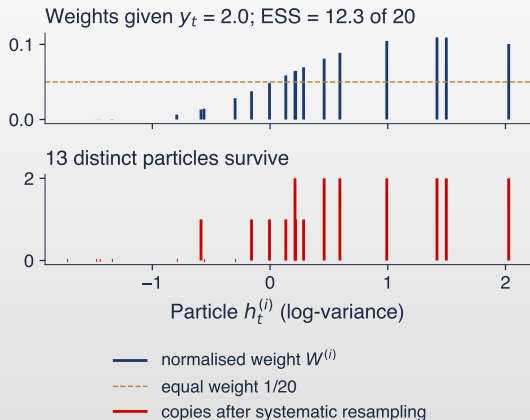
- Bootstrap particle filter [Gordon, Salmond and Smith \(1993\)](#): N particles $x_t^{(i)}$ approximate the distribution of the state
 - ▶ propagate $x_t^{(i)} \sim f(\cdot | x_{t-1}^{(i)})$ (f : transition density); weight $w^{(i)} = g(y_t | x_t^{(i)})$ (g : measurement density)
 - ▶ normalised weights $W^{(i)} = w^{(i)} / \sum_j w^{(j)}$; then resample

- Likelihood estimate and the effective sample size

$$\hat{p}(y_t | Y_{t-1}) = \frac{1}{N} \sum_{i=1}^N w^{(i)}, \quad \hat{L} = \prod_{t=1}^n \hat{p}(y_t | Y_{t-1}), \quad \text{ESS}_t = \frac{1}{\sum_i (W^{(i)})^2}$$

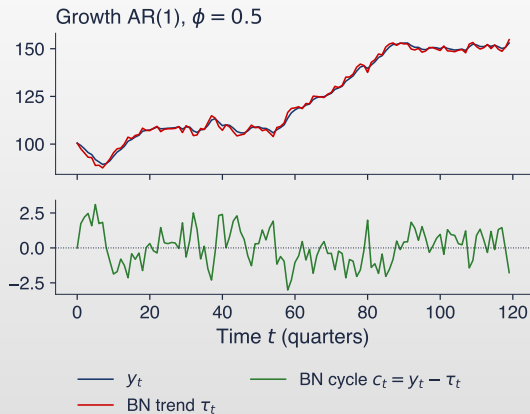
- ▶ $\hat{L}$ is unbiased: $\mathbb{E}\hat{L} = L$; by Jensen, $\mathbb{E}\ln \hat{L} < \ln L$: the log is biased downwards
 - ▶ ESS_t between 1 and N ; a collapse to a few particles: the model or the proposal fails at t
- Systematic resampling: one uniform U , $u_i = (U + i - 1)/N$
 - ▶ particle j is copied once for every $u_i \in (C_{j-1}, C_j]$; $C_j = \sum_{l \leq j} W^{(l)}$: the cumulative weights

Prerequisites for Today (15/19): one step of a particle filter



- SV model, 20 particles $h_t^{(i)}$, observation $y_t = 2.0$
 - ▶ $w^{(i)} = N(y_t; 0, e^{h_t^{(i)}})$: a large $|y_t|$ favours high-volatility particles
 - ▶ ESS = 12.3 of 20
- Systematic resampling (bottom)
 - ▶ low-volatility particles die, high ones are copied twice
 - ▶ 13 distinct particles survive
- Auxiliary filter (APF) [Pitt and Shephard \(1999\)](#): look at y_t before propagating
 - ▶ first-stage weights $\propto g(y_t | \mu_t^{(i)})$, $\mu_t^{(i)} = \mathbb{E}(x_t | x_{t-1}^{(i)})$
 - ▶ then propagate and correct the weights

Prerequisites for Today (16/19): the Beveridge–Nelson decomposition



- BN trend [Beveridge and Nelson \(1981\)](#): the level once predictable growth dies out

$$\tau_t = y_t + \sum_{j \geq 1} \mathbb{E}_t(\Delta y_{t+j} - \mu)$$

- μ : mean growth; $\mathbb{E}_t$: expectation given data up to t

- AR(1) growth: $\Delta y_t - \mu = \phi(\Delta y_{t-1} - \mu) + e_t$

- cycle $c_t = y_t - \tau_t = -\frac{\phi}{1-\phi}(\Delta y_t - \mu)$

- chart, $\phi = 0.5$: $c_t = -(\Delta y_t - \mu)$, noisy, small

- Local level: $\Delta y_t = u_t + \theta u_{t-1}$, an IMA(1,1)

- u_t : white noise; $\theta = -(1 + q/2) + \sqrt{q + q^2/4}$

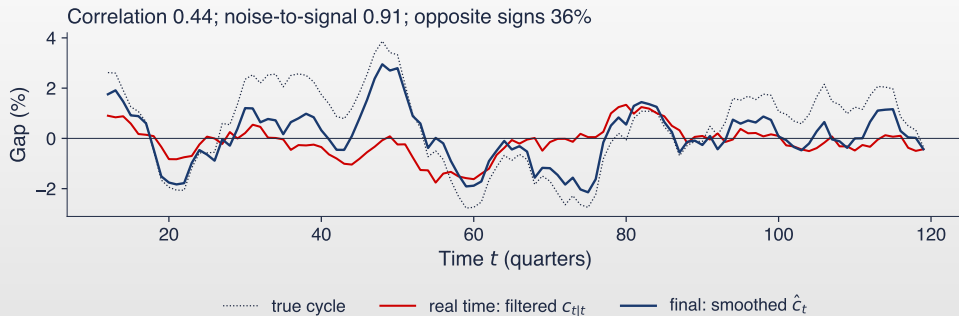
Prerequisites for Today (17/19): output gaps in real time

- Smooth-trend UC (unobserved components) model with an AR(2) cycle

$$y_t = \tau_t + c_t, \quad \Delta^2 \tau_{t+1} = \zeta_t, \quad c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \kappa_t$$

- ▶ τ_t : potential output; c_t : the output gap; ζ_t, κ_t : independent shocks; Δ^2 : second difference
- ▶ real-time gap: the filtered $c_{t|t}$ from data up to t ; final gap: the smoothed $\hat{c}_t$
- Orphanides–van Norden statistics [Orphanides and van Norden \(2002\)](#), revision $r_t = c_{t|t} - \hat{c}_t$
 - ▶ $\text{Corr}(c_{t|t}, \hat{c}_t)$; noise-to-signal ratio $\text{sd}(r_t)/\text{sd}(\hat{c}_t)$, near 1 or above: unreliable
 - ▶ share of opposite signs: $\frac{1}{n} \sum_t 1\{\text{sign}(c_{t|t}) \neq \text{sign}(\hat{c}_t)\}$

Prerequisites for Today (18/19): real-time against final gaps



- Simulated smooth trend plus AR(2) cycle ($\phi_1 = 1.4$, $\phi_2 = -0.5$), true parameters known: only the end-point problem is at work
- The filtered gap is flatter and often of the wrong sign: correlation 0.44, noise-to-signal 0.91, opposite signs in 36% of the quarters

Prerequisites for Today (19/19): trend inflation: UC-SV

- UC-SV Stock and Watson (2007): inflation = trend + transitory noise, both with stochastic volatility

$$\pi_t = \tau_t + \sigma_{\eta,t}\eta_t, \quad \tau_t = \tau_{t-1} + \sigma_{\varepsilon,t}\varepsilon_t$$

- ▶ $\eta_t, \varepsilon_t \sim N(0, 1)$, independent; τ_t : trend inflation
- ▶ $\ln \sigma_{\eta,t}^2 = \ln \sigma_{\eta,t-1}^2 + \nu_{\eta,t}$, $\ln \sigma_{\varepsilon,t}^2 = \ln \sigma_{\varepsilon,t-1}^2 + \nu_{\varepsilon,t}$, with $\text{Var}\nu = \gamma$
- ▶ γ : smoothness of the volatilities, fixed in advance (0.2 in Stock–Watson)
- Estimation by Gibbs: FFBS for τ ; KSC steps for the two log-volatilities
 - ▶ posterior probability of an event from the M draws $\tau_t^{(m)}$:
$$P(1.5 \leq \tau_t \leq 3.5 \mid Y_n) \approx \frac{1}{M} \sum_m 1\{1.5 \leq \tau_t^{(m)} \leq 3.5\}$$
 - ▶ a 68% band: the 16th and 84th percentiles of the draws

A1: the exact diffuse filter by hand [Solved]

Task

- ▣ Local level, $\sigma_\varepsilon^2 = 1$, $\sigma_\eta^2 = 0.5$, μ_1 diffuse ($a_1 = 0$, $P_{\infty,1} = 1$, $P_{*,1} = 0$); data $y = (2.0; 3.0; 2.5)$.
- ▣ 1. Run the diffuse step at $t = 1$: give F_∞ , $K^{(0)}$, $K^{(1)}$, a_2 , $P_{\infty,2}$, $P_{*,2}$.
- ▣ 2. Run the ordinary filter for $t = 2, 3$: v_t , F_t , K_t , a_{t+1} , P_{t+1} .
- ▣ 3. Compute the diffuse log-likelihood and say why y_1 carries no information about the variances.
- ▣ Report: one diffuse step, two ordinary steps, one number, one sentence.

Solution

- ▣ 1. $F_\infty = 1$, $F_* = 0 + 1 = 1$; $K^{(0)} = 1$, $K^{(1)} = 0 - 1 \cdot 1/1 = -1$;
 $a_2 = 0 + 1 \cdot 2 = 2.000$; $P_{\infty,2} = 1(1 - 1) = 0$;
 $P_{*,2} = 1 \cdot 1 + 0 + 0.5 = 1.500$
- ▣ 2. $t = 2$: $v = 1.000$, $F = 2.500$, $K = 0.600$, $a_3 = 2.600$,
 $P_3 = 1.5(1 - 0.6) + 0.5 = 1.100$; $t = 3$: $v = -0.100$,
 $F = 2.100$, $K = 0.524$, $a_4 = 2.548$, $P_4 = 1.024$
- ▣ 3. Terms -0.919 , -1.577 , -1.292 ; $\ln L_d = -3.788$. The first term is $-\frac{1}{2} \ln 2\pi$ whatever the variances: y_1 only fixes the level

A2: the concentrated likelihood [Proposed]

Solution discussed in class

Task

- ▣ The data of A1, now with $\sigma_\eta^2 = 0.5\sigma_\varepsilon^2$ and σ_ε^2 unknown. Model: A1.
- ▣ 1. Run the filter with $\sigma_\varepsilon^2 = 1$ and write $F_t = \sigma_\varepsilon^2 F_t^*$.
- ▣ 2. Derive $\hat{\sigma}_\varepsilon^2$ by maximising the diffuse log-likelihood and compute it.
- ▣ 3. Compute the concentrated log-likelihood (without the diffuse term) and state how many parameters remain for the numerical search.
- ▣ Report: two ratios, one estimate, one number, one sentence.

A3: one backward-sampling step [Solved]

Task

- Continue A1: after y_2 the filtered moments are $a_{2|2}$, $P_{2|2}$; suppose the draw for the next period is $\alpha_3 = 3.0$.
- 1. Compute $a_{2|2}$ and $P_{2|2}$ from A1.
- 2. Compute G_2 , the conditional mean m_2 and variance S_2 of $\alpha_2 | \alpha_3, y_1, y_2$.
- 3. Explain why y_3 is not needed in this conditional distribution.
- Report: two moments, three numbers, one sentence.

Solution

- 1. $a_{2|2} = 2 + 0.6 \cdot 1 = 2.600$, $P_{2|2} = 1.5 - 1.5^2/2.5 = 0.600$
- 2. $G_2 = P_{2|2}/(P_{2|2} + \sigma_\eta^2) = 0.545$;
 $m_2 = 2.600 + 0.545(3.0 - 2.600) = 2.818$;
 $S_2 = P_{2|2}(1 - G_2) = 0.273$
- 3. Markov property: given α_3 , the future observations add nothing about α_2 ; their information already entered through the draw of α_3

A4: the inverse-gamma Gibbs block [Proposed]

Solution discussed in class

Task

- ▣ A local level Gibbs sampler, $n = 50$; current state draw gives $\sum_t (y_t - \mu_t)^2 = 40$ and $\sum_t (\Delta \mu_t)^2 = 6.2$; priors $IG(2.5, 0.25)$ for both variances. Model: A3.
- ▣ 1. Write the two conditional posteriors with their parameters.
- ▣ 2. Compute their means and the implied q .
- ▣ 3. Explain why this sampler can never return $\sigma_\eta^2 = 0$, unlike ML.
- ▣ Report: two distributions, three numbers, one sentence.

A5: moments of the SV model [Solved]

Task

- ▣ SV model with $\phi = 0.98$, $\sigma_\eta = 0.15$, Gaussian ϵ_t .
- ▣ 1. Compute σ_h^2 and the kurtosis of y_t .
- ▣ 2. Compute $\text{Corr}(y_t^2, y_{t-k}^2)$ for $k = 1, 10, 50$ and the half-life of a volatility shock.
- ▣ 3. Explain the difference from GARCH(1,1) with the same persistence.
- ▣ Report: two numbers, four numbers, one sentence.

Solution

- ▣ 1. $\sigma_h^2 = 0.0225/0.0396 = 0.568$; kurtosis $3e^{0.568} = 5.30$
- ▣ 2. 0.173, 0.138, 0.054: slow decay from a low start; half-life $\ln 0.5 / \ln 0.98 = 34.3$ days
- ▣ 3. GARCH variance is known one step ahead; in SV it has its own shock, so excess kurtosis arises even with ϕ close to 1 and the likelihood needs an integral

A6: the QML measurement equation [Proposed]

Solution discussed in class

Task

- The SV model of A5, written as $\ln y_t^2 = h_t + \xi_t$. Model: A5.
- 1. Give the mean and variance of ξ_t and the signal-to-noise ratio $\sigma_h^2 / \text{Var} \xi_t$.
- 2. Compute $\text{Corr}(\ln y_t^2, \ln y_{t-k}^2)$ for $k = 1, 10$ and name the ARMA model of $\ln y_t^2$.
- 3. Explain why the Kalman filter on $\ln y_t^2$ is consistent but inefficient.
- Report: three numbers, two numbers and a model, one sentence.

A7: one step of the bootstrap particle filter [Solved]

Task

- ▣ SV model, $N = 4$ particles after propagation:
 $h^{(i)} = (-1; 0; 0.5; 1.5)$; observation $y_t = 2.0$; systematic resampling with $U = 0.3$.
- ▣ 1. Compute $w^{(i)} = N(y_t; 0, e^{h^{(i)}})$ and the normalised weights.
- ▣ 2. Compute the ESS and the likelihood increment $\hat{p}(y_t | Y_{t-1})$.
- ▣ 3. Resample: which particles survive, and how many times?
- ▣ Report: eight numbers, two numbers, one list.

Solution

- ▣ 1. $w = (0.0029; 0.0540; 0.0924; 0.1206)$;
 $W = (0.011; 0.200; 0.342; 0.447)$
- ▣ 2. ESS = 2.80 of 4; $\hat{p} = \frac{1}{4} \sum w = 0.0675$, $\ln \hat{p} = -2.696$
- ▣ 3. $u = (0.075; 0.325; 0.575; 0.825)$ against
 $C = (0.011; 0.211; 0.553; 1)$: particles 2, 3, 4, 4; the
 low-volatility particle dies, the high-volatility one is copied twice

A8: auxiliary weights with an outlier [Proposed]

Solution discussed in class

Task

- ▣ Local level, $\sigma_\varepsilon^2 = 0.5$, $\sigma_\eta^2 = 0.25$; equally weighted particles $x_{t-1}^{(i)} = (0; 1; 2; 3)$; outlier $y_t = 4.0$. Model: A7.
- ▣ 1. Compute the normalised first-stage weights $\propto N(y_t; x_{t-1}^{(i)}, \sigma_\varepsilon^2)$ and their ESS.
- ▣ 2. Compute the fully adapted weights $\propto N(y_t; x_{t-1}^{(i)}, \sigma_\varepsilon^2 + \sigma_\eta^2)$ and compare.
- ▣ 3. Explain what the bootstrap filter would do with the same outlier.
- ▣ Report: four weights and an ESS, four weights, one sentence.



A9: the Beveridge–Nelson cycle of an ARIMA(1,1,0) [Solved]

Task

- ▣ Quarterly growth: $\Delta y_t - 0.8 = 0.35(\Delta y_{t-1} - 0.8) + e_t$; this quarter $\Delta y_t = 2.0$.
- ▣ 1. Derive $\sum_{j \geq 1} \mathbb{E}_t(\Delta y_{t+j} - \mu)$ for an AR(1).
- ▣ 2. Compute the BN cycle c_t and the long-run multiplier of e_t on the trend.
- ▣ 3. Explain the sign of the cycle after a strong quarter.
- ▣ Report: one formula, two numbers, one sentence.

Solution

- ▣ 1. $\mathbb{E}_t(\Delta y_{t+j} - \mu) = \phi^j(\Delta y_t - \mu)$, sum = $\frac{\phi}{1-\phi}(\Delta y_t - \mu)$
- ▣ 2. $\frac{\phi}{1-\phi} = 0.538$; $c_t = -0.538 \cdot 1.2 = -0.646$; multiplier $1/(1-\phi) = 1.538$
- ▣ 3. With positive growth persistence, a strong quarter raises expected future growth: the trend jumps by more than output, so output is *below* trend

A10: local level, IMA(1,1) and Beveridge–Nelson [Proposed]

Solution discussed in class

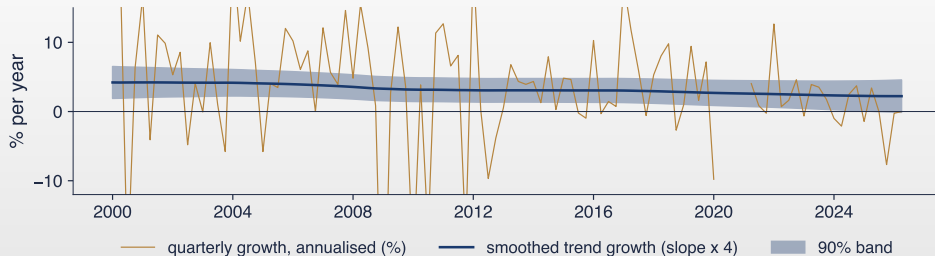
Task

- ▣ Local level with $\sigma_\varepsilon^2 = 1$, $q = 0.5$. Model: A9.
- ▣ 1. Find the MA coefficient θ of Δy_t .
- ▣ 2. Solve the steady-state Riccati equation $P = P - P^2/(P + 1) + q$ and give the gain $k = P/(P + 1)$.
- ▣ 3. Show that the BN trend of the IMA(1,1), $y_t + \theta u_t$, equals the steady-state filtered level $a_t|_t$.
- ▣ Report: one number, two numbers, one derivation.

B1: Romanian trend growth [Solved]

- ▣ **Question:** has Romania's trend growth fallen since the 2000s, once the uncertainty of the estimate is taken into account?
- ▣ **Inputs:** 100×log real GDP (Eurostat, SCA), 2000Q1–2026Q2, $T = 106$; 2020Q2–Q4 treated as missing
- ▣ **Tasks**
 1. Estimate the local linear trend by exact diffuse ML with the numpy filter and with statsmodels; compare the parameters and the log-likelihoods.
 2. Smooth the slope and report the annualised trend growth $4\hat{\nu}_t$ with a 90% band in 2005, 2012, 2019 and in the last quarter.
 3. Check whether any variance is at the boundary.
 4. Interpretation: is the fall in trend growth statistically meaningful?
- ▣ **Report:** two parameter sets, four growth rates with bands, one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



- ▣ numpy: $\sigma_\varepsilon^2 = 0.6013$, $\sigma_\xi^2 = 3.9577$, $\sigma_\zeta^2 = 0.0039$, $\ln L_d = -231.08$; statsmodels: 0.6011, 3.9581, 0.0039, -229.24 (the difference in $\ln L$ is $\ln 2\pi$, two diffuse steps)
- ▣ Trend growth: 4.1% (s.d. 1.1) in 2005, 3.1% (1.0) in 2012, 2.8% (1.1) in 2019, 2.2% (1.4) at the end
- ▣ Interpretation: a fall is suggested but not established
 - ▶ the slope variance is small, so trend growth moves slowly
 - ▶ the bands overlap widely; level shocks absorb the 2009 recession

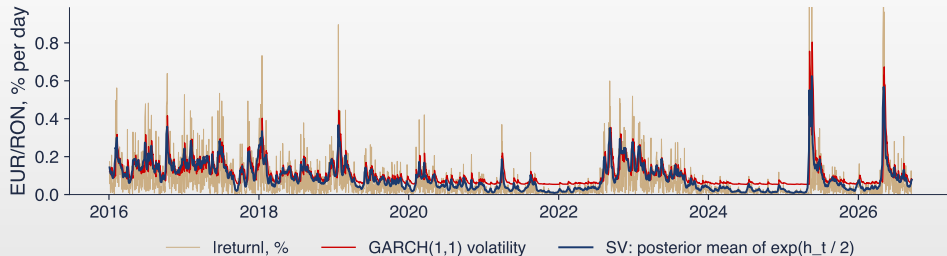
B2: TVP regressions for Hungary and Poland [Proposed]

- **Question:** has the co-movement of Hungarian and Polish inflation with euro-area inflation changed over time?
- **Inputs:** quarterly HICP inflation (% a.r.), 2005Q4 onward; euro-area inflation net of its seasonal pattern; the model of B1 extended to $y_t = c_t + b_t x_t + D_t' \gamma + \varepsilon_t$; model: B1
- **Tasks**
 1. Estimate the TVP regression by exact diffuse ML (five diffuse states) for each country.
 2. Report $\hat{\sigma}_b^2$ and the smoothed b_t with standard errors in 2008Q3, 2019Q4 and 2022Q4.
 3. Compute the LR test of $\sigma_b^2 = 0$ and its $\frac{1}{2} \chi_1^2$ p -value; say why this p -value is only indicative.
 4. Interpretation: is the slope higher than one, and is the co-movement stable or episodic?
- **Report:** a table of two rows, six slopes with standard errors, two tests, one sentence
- **Notebook:** section B2

B3: stochastic volatility of EUR/RON [Solved]

- **Question:** how persistent is the volatility of the leu against the euro, and how does SV compare with GARCH?
- **Inputs:** daily log returns of the BNR EUR/RON reference rate, %, demeaned, 2016–2026, $T = 2686$; KSC Gibbs sampler with $c = 10^{-3}\text{Var}(y)$
- **Tasks**
 1. Run the KSC Gibbs sampler and report posterior means and 90% intervals of μ, ϕ, σ_η with inefficiency factors.
 2. Estimate a Gaussian GARCH(1,1) on the same data.
 3. Plot the posterior mean of $\exp(h_t/2)$ against the GARCH volatility.
 4. Interpretation: what does the half-life of volatility say about a managed exchange rate?
- **Report:** a table of three parameters, three GARCH numbers, one chart, one sentence
- **Notebook:** section B3

B3: solution [Solved]



- ▣ $\phi = 0.957$ [0.944; 0.969], $\sigma_\eta = 0.541$ [0.488; 0.597], $\mu = -5.628$ [-6.032; -5.210]; inefficiency factors 18, 42, 1 (20 000 draws)
- ▣ GARCH(1,1): $\alpha = 0.225$, $\beta = 0.753$; daily s.d. 0.121%, kurtosis 27.8
- ▣ Interpretation: the half-life of a log-volatility shock is about 16 days
 - ▶ quiet periods with rare bursts are what a managed float produces
 - ▶ the high kurtosis says that Gaussian ϵ_t is not enough (B4)

B4: the particle filter for the EUR/RON SV model [Proposed]

- **Question:** is SV better than GARCH for EUR/RON on the likelihood scale, and is the particle likelihood reliable here?
- **Inputs:** the data and the posterior means of B3; bootstrap particle filter with systematic resampling; model: B3 and A7
- **Tasks**
 1. Run the filter with $N = 500, 2000, 10\,000$ particles, 30 times each, and report the s.d. of $\ln \hat{L}$.
 2. Report the minimum and median ESS and the date of the minimum.
 3. Compare $\ln \hat{L}$ (large N) with the GARCH maximum.
 4. Interpretation: what does the ESS collapse tell you about the model?
- **Report:** three standard deviations, two ESS values and a date, two log-likelihoods, one sentence
- **Notebook:** section B4

B5: the Romanian output gap in real time [Proposed]

- **Question:** how much does the end-of-sample estimate of the Romanian output gap change when later data arrive?
- **Inputs:** the data of B1; smooth-trend UC with AR(2) cycle (`fit_smooth_trend`); quasi-real time: re-estimate on data up to each quarter from 2012Q4 (parameters every 4 quarters) and keep the filtered gap; model: B1 and A10
- **Tasks**
 1. Compute the series of real-time (filtered, end-of-sample) gaps and the final smoothed gap.
 2. Compute the Orphanides–van Norden statistics: correlation, noise-to-signal ratio (s.d. of revisions over s.d. of the final gap), share of opposite signs.
 3. Report the largest revision.
 4. Interpretation: can monetary policy rely on the current estimate of the gap?
- **Report:** four statistics, one number, one sentence
- **Notebook:** section B5

C1: is Romanian trend inflation anchored? [Proposed]

- ▣ **Question:** is the trend of Romanian inflation inside the BNR target band ($2.5\% \pm 1$ pp), and how robust is the answer?
- ▣ **Inputs:** quarterly HICP inflation (% a.r.); UC-SV of Stock and Watson (2007) with quarterly seasonal dummies (ucsv_gibbs); model: B3
- ▣ **Tasks**
 1. Write a five-line pre-registration: hypothesis, data, the anchoring statistic, the robustness grid, what would falsify the claim.
 2. Estimate the model for start dates 2001 and 2006 and for $\gamma = 0.1$ and 0.2 ; report the trend in the last quarter with a 68% band.
 3. Compute the posterior probability that the trend lies in 1.5–3.5% in the last quarter and in 2019Q4.
 4. Interpretation: what can and cannot be concluded, and how would you turn this into a project?
- ▣ **Report:** a five-line plan, a table of four rows, a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise state space estimation. The answer:
- ▣ (a) The Kalman filter is useless if the errors are not Gaussian.
- ▣ (b) For a random-walk state, set $P_1 = 10^{-10}$ and keep all n terms of the likelihood.
- ▣ (c) An ML estimate of the level variance equal to zero proves that the level is constant.
- ▣ (d) SV and GARCH are the same model, since both make volatility persistent.
- ▣ (e) The particle-filter likelihood is unbiased, so its logarithm is unbiased too.
- ▣ (f) The Durbin-Koopman simulation smoother needs only one state-smoother pass per draw.
- ▣ (g) With an AR(1) cycle, the correlation between trend and cycle shocks can be estimated.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, a number from today's tasks (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ The exact diffuse filter lets the first observations fix the nonstationary states; they carry no information about the variances
- ▣ Sampling the states is the Kalman machinery run backwards (FFBS) or once more on simulated data (DK)
- ▣ SV adds a shock to the variance; particle filters estimate its likelihood, unbiased in levels only
- ▣ Trend growth, gaps and anchored trends are estimates with wide bands: report them, and report real-time values
- ▣ An AI answer is a draft: check the initialisation, the boundary, the log-likelihood bias and the identification

After the seminar

- Lecture 6 develops each topic of today: the exact diffuse filter, the likelihood at the boundary, the three samplers, KSC, particle filters and PMMH, TVP, DFM, MNZ and UC-SV
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: trend inflation and anchoring in Central and Eastern Europe
- Reading: [Durbin and Koopman \(2012\)](#), Ch. 4–5; [Kim, Shephard and Chib \(1998\)](#); [Morley, Nelson and Zivot \(2003\)](#); [Stock and Watson \(2007\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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